

Visitor Management System ★★★★★

Project Overview

A **Visitor Management System (VMS)** is a secure web-based application that digitizes the process of managing visitors entering an organization. Instead of maintaining paper registers, visitors can register online or at a reception kiosk, employees receive approval requests instantly, security personnel verify visitor identity using QR codes, and all visitor activities are logged for security and auditing purposes.

The system improves **security, efficiency, visitor experience, and compliance** by maintaining real-time visitor records and automating the complete check-in/check-out workflow.

Objectives

- Eliminate manual visitor registers.
 - Improve office security.
 - Enable quick visitor approvals.
 - Reduce waiting time at reception.
 - Generate digital visitor passes.
 - Track visitor movements inside the organization.
 - Maintain complete security audit logs.
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Users

1. Visitor

- Register visit details
 - Upload ID proof
 - View visit status
 - Download visitor pass
 - QR-based check-in/check-out
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2. Employee

- Receive visitor requests
- Approve or reject visitors
- Schedule meetings

- View visitor history
 - Notify reception
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3. Receptionist

- Register walk-in visitors
 - Verify visitor identity
 - Print visitor badge
 - Manage visitor queue
 - Assign meeting room
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4. Security Guard

- Scan QR code
 - Allow entry
 - Verify visitor
 - Monitor exits
 - View active visitors
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5. Admin

- Manage employees
 - Manage departments
 - Configure system
 - Generate reports
 - View security logs
 - Manage meeting rooms
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Complete Workflow

Step 1: Visitor Registration

Visitor enters:

- Name
- Mobile Number
- Email
- Company Name
- Purpose of Visit

- Department
- Employee to Meet
- Date & Time
- ID Proof Upload
- Photo Capture

System validates all information.

Step 2: Employee Approval

Employee receives notification.

Options:

- Approve
- Reject
- Request Reschedule

If approved:

- QR Code generated
 - Visitor pass created
 - Email/SMS sent
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Step 3: QR Check-in

Visitor arrives.

Reception:

- Scan QR
- Verify ID
- Capture arrival time
- Print visitor badge

Status changes:

Pending → Checked-In

Step 4: Meeting Room Allocation

Reception assigns:

- Meeting Room

- Floor
- Building
- Time Slot

System checks room availability.

Step 5: Visit Tracking

During visit:

- Meeting started
- Meeting ended
- Room occupied
- Security monitoring

Live dashboard shows:

- Active Visitors
 - Available Rooms
 - Employees in Meeting
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Step 6: QR Check-out

Visitor scans QR while leaving.

System records:

- Exit Time
- Visit Duration
- Badge Returned
- Feedback

Status:

Checked-In → Checked-Out

Main Modules

1. Visitor Module

Features

- Online Registration
- Walk-in Registration

- Visitor Pass
 - QR Generation
 - Visit History
 - ID Upload
 - Photo Capture
 - Check-in
 - Check-out
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2. Employee Approval Module

Features

- Instant Notifications
 - Accept/Reject
 - Schedule Meeting
 - View Calendar
 - Visitor History
 - Emergency Contact
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3. QR Check-in Module

Features

- QR Scanner
 - Visitor Verification
 - Identity Verification
 - Entry Recording
 - Exit Recording
 - Badge Printing
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4. Meeting Room Module

Features

- Room Booking
- Room Availability
- Meeting Schedule
- Auto Allocation
- Time Slot Management
- Room Utilization Report

5. Security Management Module

Features

- Live Visitors
 - Visitor Tracking
 - Blacklisted Visitors
 - Emergency Lockdown
 - Security Alerts
 - Watchlist Verification
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6. Reports Module

Reports

- Daily Visitors
 - Monthly Visitors
 - Visitor Trends
 - Employee Visit Report
 - Department Report
 - Room Usage Report
 - Peak Hours
 - Visitor Duration Report
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7. Notification Module

Notifications via:

- Email
- SMS
- Push Notification

Events

- Approval Request
 - Approval Status
 - Visitor Arrival
 - Meeting Reminder
 - Check-out Reminder
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Dashboard

Admin Dashboard

Cards

- Total Visitors
- Today's Visitors
- Active Visitors
- Checked-Out Visitors
- Pending Approvals
- Available Rooms
- Employees Online
- Security Alerts

Charts

- Daily Visitors
 - Monthly Visitors
 - Department-wise Visits
 - Peak Visit Hours
 - Visitor Purpose Analysis
 - Room Utilization
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Employee Dashboard

Cards

- Pending Requests
 - Approved Visitors
 - Today's Meetings
 - Upcoming Visitors
-

Reception Dashboard

Cards

- Walk-in Visitors
 - Expected Visitors
 - Active Visitors
 - Meeting Rooms
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Security Dashboard

Cards

- Visitors Inside
 - Entry Logs
 - Exit Logs
 - Suspicious Visitors
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QR Check-in System

Each approved visitor receives:

- Unique QR Code
- Visitor ID
- Name
- Employee Name
- Visit Date
- Valid Time
- Meeting Room

Reception scans QR.

System validates:

- Approval
 - Time
 - Identity
 - Visit Status
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Meeting Room Management

Features

- Add Rooms
 - Edit Rooms
 - Delete Rooms
 - Capacity
 - Projector Availability
 - Video Conference
 - Whiteboard
 - Availability Calendar
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Security Logs

Every action is stored.

Examples

Visitor Registered

Employee Approved

QR Generated

QR Scanned

Check-in Completed

Room Assigned

Meeting Started

Meeting Ended

Check-out Completed

Badge Returned

Admin Login

Employee Login

Failed Login Attempt

Visitor Rejected

Emergency Alert

These logs include:

- Timestamp
- User
- Action
- IP Address
- Device Information
- Status

Database Collections (Firebase Firestore)

visitors

- visitorId
- name
- mobile
- email
- company
- purpose
- employeeId
- department
- photoURL
- idProofURL
- qrCode
- status
- checkInTime
- checkOutTime

employees

- employeeId
- name
- department
- designation
- email
- phone

approvals

- approvalId
- visitorId
- employeeId
- status
- remarks
- approvalTime

meetingRooms

- roomId
- roomName
- capacity
- floor
- status

bookings

- bookingId

- roomId
- visitorId
- employeeId
- startTime
- endTime

securityLogs

- logId
- userId
- action
- timestamp
- ipAddress
- device
- status

notifications

- notificationId
- receiverId
- title
- message
- readStatus
- createdAt

Technology Stack

Frontend

- HTML5
- CSS3
- JavaScript
- Bootstrap 5
- Chart.js
- QR Code Library
- HTML5 QR Scanner

Backend

- Firebase Authentication
- Cloud Firestore
- Firebase Storage
- Firebase Cloud Messaging (optional)
- Firebase Hosting

Advanced Features

- Face photo capture during registration
 - Visitor blacklist management
 - Emergency evacuation visitor list
 - Multi-branch office support
 - Digital NDA/Visitor agreement signing
 - OTP verification for visitors
 - Pre-registration through invitation links
 - Host notification on visitor arrival
 - Real-time visitor tracking dashboard
 - CSV/PDF report export
 - Role-based access control (Admin, Receptionist, Security, Employee)
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Expected Outcome

The Visitor Management System provides a secure, paperless, and efficient way to manage office visitors. By integrating employee approvals, QR-based check-in/check-out, meeting room scheduling, and detailed security logs, it streamlines the entire visitor lifecycle, improves workplace security, reduces manual effort, and delivers real-time visibility into visitor activities for organizations of any size.